**School of Computer Science** UNIVERSITY OF LEEDS

# COMP5200M Project Specification

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| **Student Name:** Kaustubh Kaushal |
| **Programme of Study:** MSc. Advanced Computer Science (Data Analytics) |
| **Supervisor Name:** Prof. Roy A. Ruddle |
| **Name of External Company** (if any)**: Nil** |
| **Provisional Title of Project:**<br>Interactive Visual Analytics for Explaining and Comparing Classification Models |
| **Aim of Project:**<br>The aim of this project is to design, implement in Python, and evaluate an interactive visual analytics system that enables users to explore and compare machine learning classification models through selected Explainable AI (XAI) tasks.<br><br>Despite their widespread use, contemporary classification models frequently lack transparency. Performance metrics measure accuracy, but they don't explain how or why predictions are made. In order to support model diagnosis, analytical improvement, and model comparison (Stages 1–3 of the MAVIS framework), this project integrates interactive visualisation techniques with well-established XAI tasks. Tasks involving stakeholder communication (Stage 4) are specifically left out. |

## Objectives:

1. Conduct a systematic literature review of Explainable AI methods and visual analytics approaches for classification models, identifying limitations and scalability challenges.
2. Select and preprocess suitable mixed-type dataset(s), ensuring appropriate encoding and validation procedures.
3. Implement two classification models (e.g., Logistic Regression and Random Forest) and evaluate their performance using standard metrics (accuracy, precision, recall, F1-score, ROC-AUC).
4. Design and justify interactive visualisations addressing selected XAI tasks, including feature importance, class separation, misclassified instances, decision boundaries, and uncertainty.
5. Develop a modular prototype that allows interactive switching between models and explanation tasks.
6. Evaluate the system through analytical case studies, quantitative comparison of models, and critical discussion of limitations and future improvements.

## Deliverables:

1. MSc dissertation report
2. GitHub repository containing the implementation